Bounded Residual Reinforcement Learning over Limited LQR/IK Prior for Height-Adaptive Wheeled Biped Balance

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***Abstract*—Wheeled bipedal robots combine the efficiency of wheeled locomotion with the adaptability of articulated legs, but maintaining balance across varying body heights remains challenging. Classical model-based controllers such as LQR with height-dependent inverse kinematics provide limited structured action priors, but are insufficient as standalone solutions—our evaluation shows fall rates of 85–100% across the feasible height range (0.40–0.70 m). Pure reinforcement learning from scratch can learn robust policies but suffers from poor sample efficiency and lacks the structure that domain knowledge provides. We propose a bounded residual reinforcement learning approach that combines a height-dependent LQR/IK prior with a learned PPO residual policy. The prior provides limited structured action priors, while the bounded residual learns corrective actions for height-adaptive stabilization, commanded height transitions, and push-disturbance recovery. The residual policy receives a 52-dimensional observation that includes the 42-dimensional base observation plus the 10-dimensional base action from the prior, and outputs a bounded residual action that is composed with the base action via final $=$ clip(base $+$ scale $\times$ residual, $-1, 1$). We train the policy in MuJoCo MJX with JAX-accelerated vectorized simulation using proximal policy optimization across 4,096 parallel environments. Our evaluation of the LQR/IK prior alone shows fall rates of 85–100% across the feasible height range, confirming its role as a limited structured prior. The proposed approach will be evaluated against the LQR/IK prior alone and a pure PPO baseline trained from scratch.**

***Index Terms*—Wheeled biped robot, residual reinforcement learning, bounded residual policy, LQR/IK prior, height-adaptive balance, push recovery, MuJoCo MJX.**

I. Introduction

Wheeled biped robots combine the efficiency of wheel-based locomotion with the adaptability of legged morphology. Compared with conventional legged systems, wheeled bipeds can achieve lower-cost motion on flat terrain while exploiting articulated legs to modify body posture, regulate center-of-mass height, and improve disturbance rejection. However, these benefits come at the cost of more challenging control: balancing at different commanded heights requires managing the nonlinear coupling between wheel dynamics, torso attitude, and multi-joint leg motions.

Classical control approaches [11] can stabilize wheeled bipeds near a nominal operating point, but become increasingly inaccurate as the robot moves away from the linearization height. Recent progress in reinforcement learning (RL) has shown that robust control policies can be learned directly in simulation for complex robotic systems [14], [15], [17]. Nevertheless, training a single policy to balance across a wide range of body heights is difficult due to poor sample efficiency and the risk of reward exploitation. In particular, simple height-tracking rewards may allow policies to satisfy torso height objectives through unnatural or unstable joint configurations rather than through physically meaningful postures.

To address these challenges, we propose a bounded residual reinforcement learning approach that combines a height-dependent LQR/IK prior with a learned PPO residual policy. The prior provides limited structured action priors based on linearized dynamics and inverse kinematics, while the bounded residual learns corrective actions to compensate for modeling errors, handle height transitions, and recover from push disturbances. Our evaluation of the LQR/IK prior alone shows fall rates of 85–100% across the feasible height range (0.40–0.70 m), confirming that the prior is insufficient as a standalone controller but provides useful structure for residual learning. The residual policy receives a 52-dimensional observation that includes the base 42-dimensional proprioceptive state plus the 10-dimensional base action from the prior, enabling the policy to learn corrections that are aware of the nominal controller's intent.

The main contributions of this work are:

- A bounded residual reinforcement learning architecture that combines a height-dependent LQR/IK prior with a learned PPO residual policy for height-adaptive wheeled biped balance.
- Quantitative evaluation showing that the LQR/IK prior alone achieves 85–100% fall rates across the feasible height range (0.40–0.70 m), establishing it as a limited structured prior suitable for residual learning.
- A 52-dimensional residual observation space that includes the base 42-dimensional proprioceptive state plus the 10-dimensional base action from the prior, enabling the residual policy to learn corrections that are aware of the nominal controller's intent.
- Bounded residual action composition via final $=$ clip(base $+$ scale $\times$ residual, $-1, 1$) with per-joint residual scale vectors, preventing the residual from dominating or destabilizing the base controller.
- Evaluation protocol for random-height balance, commanded height transitions, push recovery up to 200 N, and

robustness to model uncertainty, with comparison against the LQR/IK prior alone and a pure PPO baseline.

The rest of this paper is organized as follows. Section II reviews related work. Section III presents the robot platform and simulation setup. Section IV describes the bounded residual RL architecture. Section V details the height-dependent LQR/IK prior. Section VI presents the residual policy design and training. Section VII describes the reward formulation. Section VIII details the evaluation protocol and metrics. Section IX presents experimental results. Section X discusses limitations and future work. Section XI concludes the paper.

II. Related Work

A. Classical and Model-Based Control for Wheeled Bipeds

Wheeled bipedal robots combine the energy efficiency of wheel-based locomotion with the postural adaptability of articulated legs. Early and influential work by Klemm *et al.* [10] introduced Ascento, a two-wheeled jumping robot demonstrating active balance through LQR-based stabilization around an inverted pendulum model. Subsequent work [11] extended this to an LQR-assisted whole-body controller with kinematic loops. Hsu *et al.* [12] applied fuzzy logic control to wheeled biped balance regulation. Li *et al.* [13] investigated dynamic motion control for wheel-biped transformable robots. The DIABLO platform [24] demonstrated six-DOF direct-drive wheeled biped control using cascaded LQR. While these model-based approaches provide stability guarantees near an operating point, they require accurate dynamics models and degrade when the robot deviates significantly from the linearization configuration—a limitation that becomes severe under variable-height commands.

B. Deep Reinforcement Learning for Legged Locomotion

Deep RL has produced robust locomotion policies for complex legged systems. Hwangbo *et al.* [17] introduced a learned actuator network that accurately models motor dynamics, enabling sim-to-real transfer of agile skills on ANYmal. Lee *et al.* [15] demonstrated blind proprioceptive locomotion over challenging outdoor terrain using a teacher-student distillation pipeline. Kumar *et al.* [16] proposed Rapid Motor Adaptation (RMA), a two-stage approach that trains an adaptation module online to handle unseen dynamics such as varying payload and slippery floors. Rudin *et al.* [14] showed that massively parallel GPU-based simulation (4,096 environments in Isaac Gym) can train walking policies in minutes using a terrain curriculum closely related to ours. Miki *et al.* [18] further demonstrated perception-aware locomotion over kilometer-scale outdoor terrain using attention-based proprioceptive encoders. These works establish that RL can scale to high-dimensional contact-rich control problems; our work adapts this paradigm to the specific challenge of variable-height wheeled-biped balancing.

C. Residual Reinforcement Learning

Residual RL combines model-based controllers with learned policies to leverage the benefits of both approaches. Silver *et al.* [?] introduced residual policy learning, where a learned policy outputs corrections to a fixed base controller. Johannink *et al.* [?] demonstrated residual RL for robotic manipulation, showing that learning on top of a nominal controller improves sample efficiency and final performance compared to learning from scratch. Rana *et al.* [?] proposed Bayesian residual RL for safe exploration by maintaining uncertainty estimates over the residual policy. Our work follows this paradigm: we train a bounded PPO residual policy over a height-dependent LQR/IK prior, with explicit action composition $\text{final} = \text{clip}(\text{base} + \text{scale} \times \text{residual}, -1, 1)$ that prevents the residual from dominating or destabilizing the base controller. Unlike prior work that assumes a strong base controller, we quantitatively establish that our LQR/IK prior alone achieves 85–100% fall rates across the feasible height range, confirming it as a limited but structured nominal controller suitable for residual learning.

D. Reinforcement Learning for Wheeled-Legged Robots

More recently, RL has been applied to wheeled-legged hybrid platforms. Lee *et al.* [19] presented a full autonomy stack for ANYmal-on-wheels, achieving 20 km/h driving with 83% reduced cost-of-transport. Vollenweider *et al.* [20] trained diverse locomotion skills for wheeled-legged robots through adversarial motion priors. Chamorro *et al.* [21] used asymmetric actor-critic RL with privileged terrain information for blind stair climbing on Ascento, successfully transferring to hardware for 15 cm step negotiation. Zhu *et al.* [22] introduced Whleaper, a 10-DOF wheeled biped most similar to our platform, demonstrating both LQR and PPO control for multimodal locomotion. Zhu and Hayashibe [23] applied hierarchical RL for path tracking on a wheeled biped with a sim-to-real transfer pipeline. In contrast to these platforms—which either fix body height or focus on forward locomotion—our work specifically addresses *variable-height stationary balance*, requiring the policy to operate stably across a 30 cm height range while maintaining postural integrity.

E. Curriculum Learning for Robot Control

Curriculum learning, introduced by Bengio *et al.* [9], organizes training from easy to hard examples to improve convergence and generalization. In robotics, Rudin *et al.* [14] demonstrated a terrain difficulty curriculum that substantially improves traversal capability. The Automatic Curriculum Learning (ACL) survey by Portelas *et al.* [25] systematizes curriculum strategies by learning progress, distinguishing goal-parameterized curricula from task-space curricula. Teacher algorithms such as ALP-GMM [26] generate parameterized task distributions that maximize learning progress in continuous task spaces. POET [27] co-evolves environments and agents to produce increasingly complex curricula. Our within-stage height curriculum is a monotone threshold-gated curriculum: the commanded height range expands only when a greedy evaluation pass confirms the policy has mastered the current range, directly addressing the instability that arises when a single policy must balance across configurations with qualitatively different inverted-pendulum dynamics.

F. Domain Randomization and Sim-to-Real Transfer

Sim-to-real transfer for locomotion policies relies on domain randomization to cover the reality gap. Tobin *et al.* [8] established domain randomization as a viable transfer strategy. Peng *et al.* [28] demonstrated dynamics randomization for robust quadruped locomotion on physical hardware. Tan *et al.* [29] showed that randomizing actuator parameters and adding observation noise enables Minitaur to transfer sim-trained trotting gaits to the real robot without fine-tuning. Our framework adopts these established protocols [8], [28], [29]—randomizing friction, mass properties, joint damping, and adding IMU/encoder noise and optional action delay—to prepare the trained policy for future hardware deployment.

III. Robot Platform and Simulation Setup

A. Robot Morphology

The robot considered in this work is a 10-actuated wheeled biped with five actuated joints on each leg: hip roll, hip yaw, hip pitch, knee, and wheel. The platform has a total mass of approximately $8.1\,\mathrm{kg}$, thigh length of $0.26\,\mathrm{m}$, shin length of $0.28\,\mathrm{m}$, wheel radius of $0.06\,\mathrm{m}$, and hip width of $0.23\,\mathrm{m}$. The sensing system includes one IMU and ten joint encoders. A physical robot with the same kinematic structure exists and is intended for future sim-to-real transfer, though hardware experiments are not part of the present work.

Let the joint vector be defined as

$$q_j = \begin{bmatrix} q^L_{\mathrm{hr}} & q^L_{\mathrm{hy}} & q^L_{\mathrm{hp}} & q^L_{\mathrm{kn}} & q^L_{\mathrm{wh}} & q^R_{\mathrm{hr}} & q^R_{\mathrm{hy}} & q^R_{\mathrm{hp}} & q^R_{\mathrm{kn}} & q^R_{\mathrm{wh}} \end{bmatrix}^\top, \tag{1}$$

where the superscripts L and R denote the left and right legs, respectively.

Fig. 1 illustrates the mechanical structure and joint layout. The articulated hip and knee joints enable variable-height posture regulation, while the wheel joints provide balancing capability through active torque generation.

B. Simulation Environment

All policies are trained in MuJoCo MJX [2], [3] with JAX-based [4] vectorized simulation. The physics simulation runs at $500\,\mathrm{Hz}$ with timestep

$$\Delta t_{\mathrm{sim}} = 0.002\,\mathrm{s}, \tag{2}$$

while the policy is queried at $50\,\mathrm{Hz}$,

$$\Delta t_{\pi} = 0.02\,\mathrm{s}. \tag{3}$$

Each policy action is therefore held constant for ten simulation substeps. Training is parallelized across 4096 environments on GPU [14].

Domain randomization [8] is applied during training to improve robustness. The base `balance` stage uses moderate perturbations to first learn clean stationary balance: body masses are perturbed by $\pm 10\%$, ground friction by $\pm 30\%$, and joint damping by $\pm 20\%$ relative to their nominal values. The later `balance_robust` stage widens these ranges to $\pm 15\%$, $\pm 40\%$, and $\pm 30\%$, respectively. Randomization is applied per episode at reset time. In the base `balance` stage, no external push disturbances are applied ($F_{\mathrm{push}} = 0$); random joint-position noise of $\pm 0.10\,\mathrm{rad}$ is added at reset to expose the policy to varied initial postures. Push disturbances ($40\,\mathrm{N}$ applied over 8 control steps, $\approx 0.16\,\mathrm{s}$) are introduced only in the `balance_robust` fine-tuning stage.

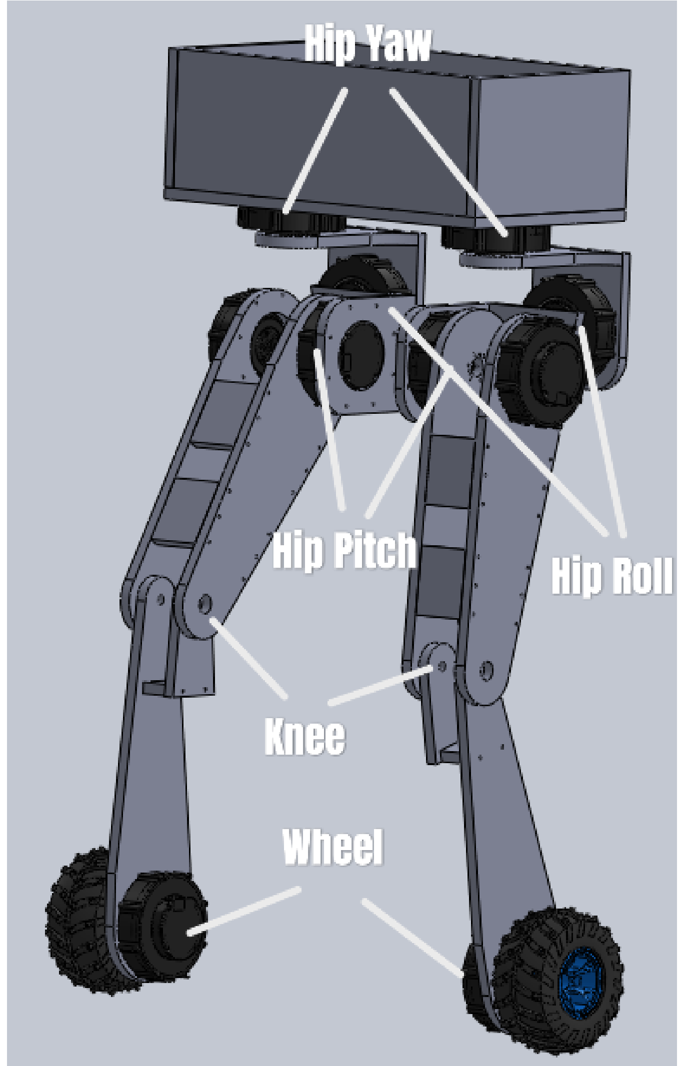

Fig. 1. Mechanical structure and joint layout of the wheeled biped robot. Each leg has five actuated joints: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). The left leg is annotated; the right leg is mirror-symmetric. The hip and knee joints enable variable-height posture regulation; the wheel joints provide balancing torque.

Table I summarizes the randomized parameters and their stage-specific ranges, following established sim-to-real protocols [8], [28]. Each parameter is sampled independently per episode from a uniform distribution over the stated range, with the nominal value at the midpoint unless noted.

TABLE I

DOMAIN RANDOMIZATION PARAMETERS APPLIED DURING TRAINING. ALL PARAMETERS ARE SAMPLED UNIFORMLY PER EPISODE. SENSOR NOISE IS SAMPLED INDEPENDENTLY PER STEP.

Parameter	Range	Unit
Dynamics		
Friction coefficient	Balance: $[0.7,\ 1.3]\times$; robust: $[0.6,\ 1.4]\times$	—
Total body mass	Balance: $[0.9,\ 1.1]\times$; robust: $[0.85,\ 1.15]\times$	—
Joint damping	Balance: $[0.8,\ 1.2]\times$; robust: $[0.7,\ 1.3]\times$	—
Sensors & Actuation		
IMU angular velocity noise	$\mathcal{N}(0,\ 0.05^2)$	rad/s
Gravity-vector noise	$\mathcal{N}(0,\ 0.02^2)$	normalized
Joint position noise	$\mathcal{N}(0,\ 0.005^2)$	rad
Joint velocity noise	$\mathcal{N}(0,\ 0.01^2)$	rad/s
Linear velocity mode	clean; optional noisy ($\sigma = 0.2$)	m/s
Action delay	0 default; optional 1 step	—
Initial Conditions		
Initial joint offset	$[-0.10,\ 0.10]$	rad

C. Low-Level PID Control

Rather than mapping policy outputs directly to actuator torques, we employ a PID low-level controller that converts normalized policy targets into torque commands. This decouples the policy from raw torque generation and simplifies the learning problem.

The policy outputs $a_t \in [-1, 1]^{10}$, which are interpreted differently depending on the joint type:

- **Leg joints** (hip roll, hip yaw, hip pitch, knee): the policy output is interpreted as a position target. A PD+I controller then computes the torque:

$$\tau_j = K_p^j(q_j^* - q_j) + K_d^j(0 - \dot{q}_j) + K_i^j \int (q_j^* - q_j)\, dt, \quad (4)$$

  where q_j^* is the position target, q_j and $\dot{q}_j$ are the measured joint position and velocity, and K_p^j, K_d^j, K_i^j are per-joint gains. The integral term includes anti-windup clamping ($\pm 0.4\,\mathrm{rad{\cdot}s}$).
- **Wheel joints**: the policy output is interpreted as a velocity target. A PI controller computes the torque:

$$\tau_w = K_p^w(\dot{q}_w^* - \dot{q}_w) + K_i^w \int (\dot{q}_w^* - \dot{q}_w)\, dt, \quad (5)$$

  where $\dot{q}_w^*$ is the velocity target. The derivative gain is masked to zero ($K_d^w = 0$) because wheel velocity-error derivative is the joint acceleration rather than joint velocity, making standard K_d semantics ill-defined for velocity-control joints.

An action-smoothing filter with coefficient $\alpha = 0.4$ blends the current and previous policy output before passing to the PID controller, reducing command chattering. Optionally, an action delay of n_{delay} control steps can model CAN-bus latency during sim-to-real preparation (default: $n_{\text{delay}} = 0$ for simulation training). All torque outputs are clipped to the actuator limits. The per-joint gains are given in Table II.

TABLE II

PER-JOINT PID GAINS. JOINT ORDER: HIP ROLL (HR), HIP YAW (HY), HIP PITCH (HP), KNEE (KN), WHEEL (WH). GAINS ARE IDENTICAL FOR LEFT AND RIGHT LEGS. THE WHEEL K_d IS MASKED TO ZERO INTERNALLY REGARDLESS OF THE TABLE VALUE.

Joint	K_p	K_i	K_d
HR	55.0	0.8	3.0
HY	40.0	0.4	2.0
HP	70.0	1.0	4.0
KN	70.0	1.0	4.0
WH	4.0	0.1	(0.0)

IV. BOUNDED RESIDUAL RL ARCHITECTURE

A. Overview

Our approach combines a height-dependent LQR/IK prior with a learned PPO residual policy. The prior provides limited structured action priors based on linearized dynamics and inverse kinematics, while the bounded residual learns corrective actions.

The control pipeline is:

$$\begin{aligned} a_t^{\text{base}} &= \pi_{\text{LQR/IK}}(o_t^{\text{base}}, h_t^{\text{cmd}}), \\ a_t^{\text{res}} &= \pi_{\text{PPO}}(o_t^{\text{res}}), \\ a_t^{\text{final}} &= \text{clip}(a_t^{\text{base}} + s \odot a_t^{\text{res}}, -1, 1), \end{aligned} \quad (6)$$

where $o_t^{\text{base}} \in \mathbb{R}^{42}$ is the base observation, $o_t^{\text{res}} = [o_t^{\text{base}}, a_t^{\text{base}}] \in \mathbb{R}^{52}$ is the residual observation that includes the base action, $s \in \mathbb{R}^{10}$ is the per-joint residual scale vector, and $\odot$ denotes element-wise multiplication.

The final action a_t^{final} is passed through action smoothing, optional delay, and the PID low-level controller described in Section III-C.

B. Base Observation Space

The base observation $o_t^{\text{base}} \in \mathbb{R}^{42}$ is shared by both the LQR/IK prior and the residual policy:

$$o_t^{\text{base}} = \left[g_t^b,\ v_t^b,\ \omega_t^b,\ q_{j,t},\ \dot{q}_{j,t},\ a_{t-1}^{\text{final}},\ \bar{h}_t^{cmd},\ \bar{z}_t,\ e_{\psi,t}\right], \quad (7)$$

where:

- $g_t^b \in \mathbb{R}^3$: gravity vector expressed in the body frame (from IMU accelerometer);
- $v_t^b \in \mathbb{R}^3$: body-frame linear velocity;
- $\omega_t^b \in \mathbb{R}^3$: body-frame angular velocity;
- $q_{j,t} \in \mathbb{R}^{10}$: joint positions;
- $\dot{q}_{j,t} \in \mathbb{R}^{10}$: joint velocities;
- $a_{t-1} \in \mathbb{R}^{10}$: previous smoothed policy action;
- $\bar{h}_t^{cmd} \in \mathbb{R}$: normalized height command;
- $\bar{z}_t \in \mathbb{R}$: normalized current torso height;
- $e_{\psi,t} \in \mathbb{R}$: yaw error (heading drift from episode start).

C. Residual Observation Space

For the bounded residual policy, the observation is extended to include the nominal prior's action:

$$o_t^{\text{res}} = \left[o_t^{\text{base}},\ a_t^{\text{base}}\right] \in \mathbb{R}^{52}, \quad (8)$$

where $o_t^{\text{base}} \in \mathbb{R}^{42}$ is defined in Eq. (7) and $a_t^{\text{base}} \in [-1, 1]^{10}$ is the normalized action output from the LQR/IK prior (Section VII).

Appending the base action to the observation allows the residual policy to condition its correction on the nominal controller's intent. This design enables the policy to learn when to trust the prior (outputting near-zero residuals) and when to override it (outputting larger corrections).

Observations are normalized online using Welford's running mean and variance estimator [7], updated during training rollouts and held fixed during evaluation.

D. Action Composition

The residual policy outputs a bounded correction $a_t^{\text{res}} = \pi_\theta(o_t^{\text{res}}) \in [-1, 1]^{10}$. The final action sent to the low-level controller is:

$$a_t^{\text{final}} = \text{clip}\big(a_t^{\text{base}} + s \odot a_t^{\text{res}},\ -1,\ 1\big), \tag{9}$$

where $s \in \mathbb{R}^{10}$ is a per-joint residual scale vector and $\odot$ denotes element-wise multiplication.

The residual scale s controls the maximum correction magnitude per joint. We use joint-specific scales to reflect the varying authority needed across the kinematic chain:

$$s = \begin{bmatrix} 0.10 & 0.05 & 0.15 & 0.15 & 0.30 \\ 0.10 & 0.05 & 0.15 & 0.15 & 0.30 \end{bmatrix}^\top, \tag{10}$$

corresponding to left and right {hip_roll, hip_yaw, hip_pitch, knee, wheel}.

Wheels receive larger residual authority ($s = 0.30$) because rapid wheel adjustments are critical for balance recovery under push disturbances. Hip pitch and knee receive moderate authority ($s = 0.15$) to refine the IK-based posture. Hip yaw is kept small ($s = 0.05$) to prevent excessive twisting, and hip roll receives moderate authority ($s = 0.10$) for lateral stabilization.

The final action a_t^{final} is converted to actuator torques through the low-level PID controller described in Section III-C.

E. Learning Objective

We seek a residual policy π_θ that maximizes the expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma^t r_t\right], \tag{11}$$

where $\gamma = 0.99$ and r_t is the reward at step t.

The policy π_θ maps the residual observation $o_t^{\text{res}} \in \mathbb{R}^{52}$ to a bounded residual action $a_t^{\text{res}} \in [-1, 1]^{10}$. The final action sent to the robot is computed via Eq. (9).

Training is performed using Proximal Policy Optimization (PPO) [1] with clipped surrogate objective

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t\Big[\min\Big(\rho_t(\theta)\hat{A}_t,\ \text{clip}\big(\rho_t(\theta), 1-\epsilon, 1+\epsilon\big)\hat{A}_t\Big)\Big], \tag{12}$$

where $\rho_t(\theta) = \pi_\theta(a_t^{\text{res}}|o_t^{\text{res}})/\pi_{\theta_{\text{old}}}(a_t^{\text{res}}|o_t^{\text{res}})$ and $\epsilon = 0.20$. The full objective adds a clipped value loss (coefficient 1.0) and entropy regularization (coefficient 0.005). Advantages $\hat{A}_t$ are computed using Generalized Advantage Estimation (GAE) [6] with $\lambda = 0.95$.

TABLE III
REWARD COMPONENTS AND WEIGHTS FOR THE VARIABLE-HEIGHT `BALANCE` TASK. WEIGHTS FOR THE `BALANCE_ROBUST` FINE-TUNING STAGE DIFFER: $w_{\text{still}} = 0$, $w_{\text{wh}} = 0$, $w_{\text{nat}} = 1.5$, $w_{\text{alive}} = 0.5$, $w_{\text{drift}} = 1.0$, AND $w_{\Delta a} = -0.005$, TO ACCOMMODATE WHEEL-BASED PUSH RECOVERY MOTION.

Component	Type	Weight
Height tracking (r_{height})	Positive	2.5
Body level (r_{level})	Positive	1.5
Position drift (r_{drift})	Positive	2.5
Orientation (r_{orient})	Positive	1.0
Symmetry (r_{sym})	Positive	1.0
Heading (r_{heading})	Positive	0.5
Legs forward (r_{fwd})	Positive	0.5
Legs vertical (r_{vert})	Positive	0.5
No motion (r_{still})	Positive	1.0
Yaw suppression (r_{yaw})	Positive	0.5
Natural pose (r_{nat})	Positive	0.4
Alive bonus (r_{alive})	Positive	0.3
Action rate ($p_{\Delta a}$)	Penalty	-0.08
Wheel velocity (p_{wh})	Penalty	-0.006
Joint torque (p_τ)	Penalty	-0.0008
Joint velocity ($p_{\dot{q}}$)	Penalty	-0.001

V. REWARD DESIGN

A. Reward Structure

The total reward at each step is a weighted sum:

$$r_t = \sum_i w_i\, r_t^{(i)}. \tag{13}$$

The complete set of reward terms and weights for the `balance` task are given in Table III.

B. Primary Reward Terms

1) Body-Level Reward: To encourage the torso to remain parallel to the ground:

$$r_{\text{level}} = \exp\left(-\frac{\phi_t^2}{\sigma_r^2}\right)\exp\left(-\frac{\theta_t^2}{\sigma_p^2}\right), \tag{14}$$

where ϕ_t and θ_t are the torso roll and pitch angles, and $\sigma_r = \sigma_p = 0.15\,\text{rad}$. The threshold $\sigma = 0.15\,\text{rad}$ ($\approx 8.6^\circ$) corresponds to the boundary of stable balance for a wheeled inverted pendulum at the nominal height: beyond this lean angle the TWIP linearization becomes inaccurate and recovery is uncertain. This choice ensures the reward remains near 1.0 during upright stance and falls sharply only when tilt enters the unstable regime, providing a tight but achievable target for the policy.

2) Height Tracking Reward:

$$r_{\text{height}} = \exp\left(-\frac{(h_t - h_t^{cmd})^2}{\sigma_h^2}\right), \tag{15}$$

with $\sigma_h = 0.25\,\text{m}$. This bandwidth was selected to provide a non-vanishing gradient signal across the full commanded height range $[0.40, 0.70]\,\mathrm{m}$: at the midpoint of a $0.15\,\mathrm{m}$ tracking error, the reward retains $\exp(-0.15^2/0.25^2) \approx 0.83$ of its maximum value, ensuring a smooth gradient that guides learning even during early training when height tracking is poor. A tighter σ would produce near-zero gradients for heights far from the target and destabilize early-stage curriculum learning.

3) Heading and Position Regularization: To keep the robot near its initial yaw ψ_0 and anchor position $p_{xy,0}$:

$$r_{\text{heading}} = \exp\left(-\frac{\mathrm{wrap}(\psi_t - \psi_0)^2}{\sigma_\psi^2}\right), \tag{16}$$

$$r_{\text{drift}} = \exp\left(-\frac{\|p_{xy,t} - p_{xy,0}\|_2^2}{\sigma_d^2}\right), \tag{17}$$

with $\sigma_\psi = 0.5\,\mathrm{rad}$. For the `balance` stage we set $\sigma_d = 0.25\,\mathrm{m}$ to penalize rolling drift early; recovery-oriented stages may use a wider drift scale.

4) Orientation and Yaw Stability: Linear orientation stability (avoids saturation at high angular rates):

$$r_{\text{orient}} = \mathrm{clip}\big(1 - 0.15\,\|\omega_t^b\|_2,\ 0,\ 1\big), \tag{18}$$

and a dedicated yaw suppression term:

$$r_{\text{yaw}} = \mathrm{clip}\big(1 - 0.5\,|\omega_{z,t}^b|,\ 0,\ 1\big). \tag{19}$$

5) Leg Symmetry:

$$r_{\text{sym}} = \exp\left(-\frac{\left\|q_{\text{leg},t}^L - q_{\text{leg},t}^R\right\|_2^2}{\sigma_{\text{sym}}^2}\right), \tag{20}$$

where $q_{\text{leg}}^{L/R}$ includes hip roll, hip yaw, hip pitch, and knee for each leg, and $\sigma_{\text{sym}} = 0.5$.

6) Leg Alignment Rewards: Two terms encourage well-aligned leg orientation:

Legs forward (r_{fwd}): rewards hip yaw close to zero (feet pointing forward):

$$r_{\text{fwd}} = \exp\left(-\frac{(q_{\text{hy}}^L)^2}{\sigma_\ell^2}\right)\exp\left(-\frac{(q_{\text{hy}}^R)^2}{\sigma_\ell^2}\right). \tag{21}$$

Legs vertical (r_{vert}): rewards hip roll close to zero (legs perpendicular to the ground plane):

$$r_{\text{vert}} = \exp\left(-\frac{(q_{\text{hr}}^L)^2}{\sigma_\ell^2}\right)\exp\left(-\frac{(q_{\text{hr}}^R)^2}{\sigma_\ell^2}\right), \tag{22}$$

where $\sigma_\ell = 0.15\,\mathrm{rad}$ for both terms.

7) No-Motion Reward: To encourage stationary balance:

$$r_{\text{still}} = \exp\left(-\frac{\|v_t^b\|_2^2}{\sigma_v^2}\right), \tag{23}$$

The velocity scale is $\sigma_v = 0.2\,\mathrm{m/s}$. This term is set to zero in the `balance_robust` stage, where wheel motion is needed for push recovery.

C. Height-Consistent Natural Pose Reward

A key design of our method is a posture regularization term that enforces consistency between the commanded torso height and the desired leg configuration. Let the desired hip-pitch and knee angles $q_{\text{hp}}^*(h^{cmd})$ and $q_{\text{kn}}^*(h^{cmd})$ be computed by piecewise-linear interpolation.

The commanded height range used by the balance-family policies is $h^{cmd} \in [0.40, 0.70]\,\mathrm{m}$. The interpolation map below uses slightly wider kinematic anchors to cover reset perturbations and mechanical limits; the anchor values $0.38\,\mathrm{m}$ and $0.72\,\mathrm{m}$ are not additional sampled command levels. Let h denote the height supplied to this interpolation map ($h = h^{cmd}$ during normal command tracking).

The kinematic keypoints, derived from forward kinematics of the two-link leg model, are:

$$\begin{aligned} h = 0.72\,\mathrm{m}:&\quad q_{\text{hp}}^* = 0.0, \quad q_{\text{kn}}^* = 0.0 \\ h = 0.71\,\mathrm{m}:&\quad q_{\text{hp}}^* = 0.3, \quad q_{\text{kn}}^* = 0.5 \\ h = 0.38\,\mathrm{m}:&\quad q_{\text{hp}}^* = 1.5, \quad q_{\text{kn}}^* = 2.5 \end{aligned}$$

For the high-standing anchor segment $h \in [0.71, 0.72]\,\mathrm{m}$:

$$t_h = \frac{h - 0.71}{0.01}, \quad q_{\text{hp}}^* = 0.3(1 - t_h), \quad q_{\text{kn}}^* = 0.5(1 - t_h). \tag{24}$$

For the lower-height anchor segment $h \in [0.38, 0.71]\,\mathrm{m}$ (which contains the full commanded training range):

$$t_l = \frac{h - 0.38}{0.33}, \quad q_{\text{hp}}^* = 1.5 - 1.2\,t_l, \quad q_{\text{kn}}^* = 2.5 - 2.0\,t_l. \tag{25}$$

The posture error is

$$e_{\text{pose}} = \Big((q_{\text{hp}}^L - q_{\text{hp}}^*)^2 + (q_{\text{hp}}^R - q_{\text{hp}}^*)^2 + (q_{\text{kn}}^L - q_{\text{kn}}^*)^2 + (q_{\text{kn}}^R - q_{\text{kn}}^*)^2\Big)^{1/2}. \tag{26}$$

The natural pose reward is

$$r_{\text{nat}} = \exp\left(-\frac{e_{\text{pose}}^2}{\sigma_{nat}^2}\right) r_{\text{knee}}, \tag{27}$$

where $\sigma_{\text{nat}} = 0.5\,\mathrm{rad}$ and r_{knee} penalizes knee hyperextension:

$$r_{\text{knee}} = \exp\left(-\frac{\min(q_{\text{kn}}^L, 0)^2}{\sigma_k^2}\right)\exp\left(-\frac{\min(q_{\text{kn}}^R, 0)^2}{\sigma_k^2}\right), \tag{28}$$

with $\sigma_k = 0.15\,\mathrm{rad}$.

This reward prevents the policy from exploiting unnatural joint configurations (e.g., extreme hip abduction with nearly straight knees) to satisfy a height target without a kinematically meaningful posture.

Design rationale. Without this term, a height-tracking reward alone permits degenerate solutions: the robot can achieve the target CoM height by bending one leg more than the other (asymmetric collapse), hyperextending the knee, or adopting an extreme hip-pitch that satisfies the height constraint but produces an unstable inverted-pendulum configuration. The piecewise-linear interpolation maps each commanded height to a physically meaningful (hip-pitch, knee) pair derived from the two-link leg geometry, effectively constraining the null space of the height objective to stable, symmetric postures. The soft exponential penalty ($\sigma_{\text{nat}} = 0.5\,\text{rad}$) allows exploration during early training while strongly penalizing configurations that deviate more than $\approx 0.5\,\text{rad}$ from the target posture.

D. Effort and Smoothness Penalties

We regularize torque magnitude, joint velocity, final action variation, and wheel rotation:

$$p_{\tau} = \sum_{j=1}^{10} \tau_{t,j}^2, \quad p_{\dot{q}} = \sum_{j=1}^{10} \dot{q}_{j,t}^2, \tag{29}$$

$$p_{\Delta a^{\text{final}}} = \sum_{j=1}^{10} (a_{t,j}^{\text{final}} - a_{t-1,j}^{\text{final}})^2, \tag{30}$$

$$p_{\text{wh}} = \dot{q}_{\text{wh},t}^{L\,2} + \dot{q}_{\text{wh},t}^{R\,2}. \tag{31}$$

The action rate penalty applies to the final composed action a_t^{final} (Eq. (9)), not the raw residual output, to penalize high-frequency corrections in the actual control signal sent to the robot.

The wheel velocity penalty and r_{still} together enforce stationary balance in the `balance` stage; both are set to zero in `balance_robust`.

E. Residual-Specific Regularization

For the bounded residual policy, we add three regularization terms to encourage the policy to rely on the nominal prior when possible and only apply corrections when necessary.

1) Residual Magnitude Penalty:

$$p_{\text{res}} = \sum_{j=1}^{10} (a_{t,j}^{\text{res}})^2, \tag{32}$$

penalizes the squared magnitude of the residual action. This encourages the policy to output near-zero residuals when the base controller is sufficient, reducing unnecessary corrections.

2) Residual Rate Penalty:

$$p_{\Delta a^{\text{res}}} = \sum_{j=1}^{10} (a_{t,j}^{\text{res}} - a_{t-1,j}^{\text{res}})^2, \tag{33}$$

penalizes high-frequency changes in the residual output, promoting smooth corrections over time.

3) Residual Saturation Penalty:

$$p_{\text{sat}} = \frac{1}{10} \sum_{j=1}^{10} \mathbb{1}\left[|a_{t,j}^{\text{res}}| > 0.95\right], \tag{34}$$

penalizes residual actions near the $[-1, 1]$ bounds. Saturation indicates the policy is attempting corrections beyond the allowed residual scale, which may signal insufficient base controller authority or overly aggressive residual behavior.

The weights for these residual penalties are: $w_{\text{res}} = -0.01$, $w_{\Delta a^{\text{res}}} = -0.005$, $w_{\text{sat}} = -0.10$.

F. Termination Conditions

An episode terminates early if the torso tilt exceeds $0.8\,\text{rad}$ ($\approx 46°$) or the torso height drops below $0.3\,\text{m}$. Otherwise, episodes run for 1000 steps ($\approx 20\,\text{s}$). A survival bonus $r_{\text{alive}} \in \{0, 1\}$ is given at each non-terminal step, providing a constant per-step incentive to remain standing.

VI. Curriculum Learning Strategy

A. Within-Stage Height Curriculum

Rather than sampling the commanded height from the full feasible range from the beginning, we employ a within-stage curriculum [9] that progressively expands the lower bound of the height command distribution.

The height lower bound is decreased by a fixed step δ_h only when all curriculum gate conditions pass:

$$h_{\min}^{(k+1)} = \max\big(h_{\min}^{(k)} - \delta_h,\ h_{\min}^{\text{final}}\big), \quad \text{if } \mathcal{G}_{\text{curr}} = 1. \tag{35}$$

Advancement signal. Every 10^7 environment steps, corresponding to 77 PPO updates under the 4096-environment, 32-step rollout configuration, the current policy is evaluated greedily (no exploration noise) using a JIT-compiled fixed-horizon rollout with 256 parallel evaluation environments. Each environment contributes one complete 1000-step episode, and the observation normalizer is held fixed. The per-step return is

$$\bar{r}_{\text{eval}} = \frac{R_{\text{mean}}}{T}, \tag{36}$$

where R_{mean} is the mean episode return and $T = 1000$ is the episode length. Advancement fires only when all gates pass:

$$\begin{aligned} &\bar{r}_{\text{train}} \geq 7.0, \\ &\bar{r}_{\text{eval}} \geq 0.70\, r_{\max}, \\ &s_{\text{eval}} \geq 0.99, \\ &f_{\text{eval}} \leq 0.01. \end{aligned} \tag{37}$$

Here $\mathcal{G}_{\text{curr}} = 1$ denotes that the conjunction in Eq. (37) holds. where s_{eval} and f_{eval} are the evaluation success and fall rates. The sum of positive `balance` reward weights is $r_{\max} = 12.2$, so the per-step evaluation threshold is $r_{\text{thr}} = 0.70 r_{\max} = 8.54$ reward/step. Using a greedy evaluation pass rather than the noisy per-update training reward produces a more reliable signal for curriculum decisions.

Curriculum parameters. Initial lower bound: $h_{\min}^{\text{init}} = 0.69\,\text{m}$; final lower bound: $h_{\min}^{\text{final}} = 0.40\,\text{m}$; $N_c = 29$ discrete levels; $\delta_h \approx 0.01\,\text{m}$; threshold ratio 0.70 ($r_{\text{thr}} = 8.54$ reward/step), with the additional train-reward, success-rate, and fall-rate gates above. At each level k:

$$h_t^{cmd} \sim \mathcal{U}\big(h_{\min}^{(k)},\ h_{\max}\big), \tag{38}$$

where $h_{\max} = 0.70\,\text{m}$.

The 29 levels span three qualitative phases:

- **Phase A** (levels 1–5): $h_{\min} \in [0.65, 0.69]\,\text{m}$; narrow band near nominal; policy focuses on stable upright stance.
- **Phase B** (levels 6–20): $h_{\min} \in [0.50, 0.64]\,\text{m}$; widening toward moderate crouching.
- **Phase C** (levels 21–29): $h_{\min} \in [0.40, 0.49]\,\text{m}$; full operating range including deep crouching.

B. Multi-Stage Macro-Curriculum

A multi-stage curriculum manager chains training stages with warm-starting and performance-gated promotion and demotion. The three stages are:

1) **`balance`** (this paper): variable-height stationary balance with within-stage height curriculum as described above. No external push disturbances ($F_{\text{push}} = 0$). Macro-curriculum promotion threshold: 80% success rate with $w_{\text{eval}} = 7.0\,\text{reward/step}$.
2) **`stand_up`**: height transitions over the same commanded range $h^{cmd} \in [0.40, 0.70]\,\text{m}$ and recovery from fallen or non-nominal initial poses. This stage keeps the same 42-dimensional observation and PID action interface as `balance`, enabling direct checkpoint transfer without changing the policy architecture. Initialized (warm-started) from the `balance` checkpoint. Promotion threshold: 80% success rate with $w_{\text{eval}} = 5.0\,\text{reward/step}$.
3) **`balance_robust`**: fine-tuning under periodic random push disturbances ($F_{\text{push}} = 40\,\text{N}$, applied over 8 control steps, every 200 steps $\approx 4\,\text{s}$). It preserves the same observation and low-level action interface and is initialized from the `stand_up` checkpoint. Reward weights modified: $w_{\text{still}} = w_{\text{wh}} = 0$; w_{nat} increased to 1.5; $w_{\Delta a}$ reduced to -0.005. Promotion threshold: 80% success rate with $w_{\text{eval}} = 6.0\,\text{reward/step}$.

Demotion fires when success rate falls below 30% within a window of 100 evaluation episodes. Stages `stand_up` and `balance_robust` have corresponding environment implementations and training configurations but have not yet been trained; their results are not reported here.

VII. Height-Dependent LQR/IK Prior

A. Overview

The nominal prior controller provides structured balance and posture commands that serve as the base action a_t^{base} in Eq. (9). The prior combines a linearized two-wheeled inverted pendulum (TWIP) LQR controller for sagittal balance with height-dependent inverse kinematics (IK) for leg posture and independent PD controllers for lateral and yaw stabilization.

The prior operates on the same 42-dimensional observation interface and shares the same PID low-level controller as the residual policy, enabling direct composition without actuator-interface mismatches.

B. Linearized TWIP Model

The sagittal balance component is based on a linearized two-wheeled inverted pendulum (TWIP) model. The state and input are

$$x_{\text{lqr}} = \begin{bmatrix} \theta_{\text{pitch}} \\ \dot{\theta}_{\text{pitch}} \\ v_{\text{fwd}} \\ p_{\text{fwd}} \end{bmatrix}, \quad u = \bar{\omega}_{\text{wheel}}, \tag{39}$$

where θ_{pitch} is the forward lean angle, v_{fwd} is the forward body velocity, p_{fwd} is the forward position drift, and u is the average wheel velocity command.

The continuous-time linearization $\dot{x}_{\text{lqr}} = A\,x_{\text{lqr}} + B\,u$ uses robot constants: mass $m = 8.1\,\text{kg}$, nominal CoM height $l_c = 0.54\,\text{m}$ (updated per height command via IK), and wheel radius $r_w = 0.06\,\text{m}$.

The LQR gain is solved from the continuous algebraic Riccati equation (CARE) with state cost $Q = \text{diag}(10, 2, 3, 0.3)$ and input cost $R = 0.8$:

$$K_{\text{lqr}} = R^{-1}B^{\top}P, \quad A^{\top}P + PA - PBR^{-1}B^{\top}P + Q = 0. \tag{40}$$

C. Height-Dependent Inverse Kinematics

To extend the LQR to variable commanded heights, we perform an offline IK scan over hip-pitch angles from 0 to $1.35\,\text{rad}$, with the knee set to $q_{\text{kn}} = 2\,q_{\text{hp}}$ (symmetric two-link fold geometry). The resulting height–angle mapping is fit by a degree-2 polynomial, yielding target joint angles for any commanded height in $[0.40, 0.70]\,\text{m}$. The effective CoM height $l_c(h^{cmd})$ fed into the TWIP linearization is updated accordingly.

D. Multi-Axis Control

Lateral and yaw axes are controlled by independent PD terms:

$$u_{\text{lat}} = K_{p,\text{roll}} \cdot e_{\text{roll}} + K_{d,\text{roll}} \cdot \dot{e}_{\text{roll}}, \tag{41}$$

$$u_{\text{yaw}} = K_{p,\text{yaw}} \cdot e_{\psi} + K_{d,\text{yaw}} \cdot \omega_z, \tag{42}$$

with $K_{p,\text{roll}} = 0.4$, $K_{d,\text{roll}} = 0.08$, $K_{p,\text{yaw}} = 2.5$, $K_{d,\text{yaw}} = 0.2$. The final wheel command is the superposition of sagittal LQR, lateral PD, and yaw PD outputs, clipped to $\pm 20.0\,\text{rad/s}$.

E. Limitations and Motivation for Residual Learning

The LQR/IK prior has several known limitations that motivate the bounded residual learning approach:

1) **Linear regime only.** The TWIP linearization is valid near the nominal standing height and small lean angles ($|\theta_{\text{pitch}}| \lesssim 10°$). At large lean deviations or very low commanded heights, accuracy degrades.
2) **No lean integral.** Without an integral term on lean error, constant disturbances (friction variation, slopes) produce nonzero steady-state lean.
3) **No large-disturbance recovery.** Under strong pushes the state leaves the linear regime; the LQR provides no recovery mechanism.
4) **Decoupled axes.** Sagittal, lateral, and yaw dynamics are treated independently, ignoring cross-coupling during aggressive maneuvers.

Empirical validation. Evaluation of the LQR/IK prior alone (Section IX-A) shows fall rates of 85–100% across the commanded height range $[0.40, 0.70]\,\text{m}$, confirming that the prior is insufficient for robust height-adaptive balance without learned corrections.

These limitations motivate the bounded residual approach: the prior provides structured balance and posture commands that are approximately correct, while the residual policy learns corrective actions to handle nonlinearities, disturbances, and cross-axis coupling.

VIII. Residual Policy Training

A. Network Architecture

The residual policy uses separate MLPs for policy and value function: hidden sizes $\{256, 256, 128\}$, ELU activation, orthogonal weight initialization, implemented in Flax [5]. The policy network takes the 52-dimensional residual observation o_t^{res} (Eq. (8)) as input and outputs the mean of a Gaussian action distribution with a learned log-standard-deviation initialized to $\log(0.2)$.

The value network shares the same architecture and observation input, estimating the expected return from the current state.

B. Training Configuration

The residual policy is trained using PPO with 4096 parallel environments, rollout length 32 steps, 4 epochs per update, 4 minibatches, learning rate 3×10^{-4} (Adam), gradient norm clipping 0.5, entropy coefficient 0.005, value-loss coefficient 1.0. To avoid rare destructive policy jumps during long resumed runs, an update is rejected if its approximate policy KL exceeds 0.03.

At each environment step, the LQR/IK prior computes the base action a_t^{base} from the current observation, which is then appended to form the 52-dimensional residual observation o_t^{res}. The policy outputs the residual action a_t^{res}, which is composed with the base action via Eq. (9) to produce the final action sent to the low-level controller.

Long training runs use a single NVIDIA Tesla V100-SXM2 GPU with 32 GB memory, an Intel Xeon Gold 6248 CPU at 2.50 GHz, and 94 GB of system RAM.

The complete set of hyperparameters is summarized in Table IV for reproducibility.

TABLE IV
PPO TRAINING HYPERPARAMETERS FOR THE BALANCE STAGE.

Hyperparameter	Value
Environment	
Training GPU	NVIDIA Tesla V100-SXM2, 32 GB
CPU / system RAM	Intel Xeon Gold 6248, 2.50 GHz / 94 GB
Parallel environments	4,096
Rollout length	32 steps
Control frequency	50 Hz (20 ms/step)
Episode length	1,000 steps (20 s)
PPO Update	
Epochs per update	4
Minibatches per epoch	4
Minibatch size	32,768
PPO clip range (ϵ)	0.20
Approx. KL reject threshold	0.03
Optimization	
Optimizer	Adam
Learning rate	3×10^{-4}
Gradient norm clip	0.5
Value loss coefficient	1.0
Entropy coefficient	0.005
Advantage Estimation	
Discount γ	0.99
GAE λ	0.95
Network	
Hidden layers (actor)	$\{256, 256, 128\}$
Hidden layers (critic)	$\{256, 256, 128\}$
Activation	ELU
Weight init	Orthogonal
Initial log-std	$\log(0.2)$
Curriculum Evaluation	
Eval interval	every 10^7 env steps (≈ 77 updates)
Eval environments	256
Eval episodes	256

C. Research Evaluation Protocol

Research evaluation is performed offline using a deterministic single-environment MuJoCo rollout with per-step telemetry recording. This evaluation pipeline is strictly separate from the in-training curriculum advancement evaluation, which uses a vectorized greedy rollout over the current policy to gate height-range expansion.

Evaluation scenarios. Ten scenario groups are defined:

1) **Nominal**: flat floor, $h^{cmd} = 0.65\,\mathrm{m}$, standard domain randomization.
2) **Fixed-height sweep**: evaluate at seven fixed heights $h^{cmd} \in \{0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70\}\,\mathrm{m}$ to assess height-adaptive performance.
3) **Random-height balance**: $h^{cmd} \sim \mathcal{U}[0.40, 0.70]$ (full commanded range).
4) **Commanded height transitions**: step changes in commanded height to assess standing-squatting transitions.
5) **Push recovery**: single impulse at $t = 3\,\mathrm{s}$; measures recovery time and maximum recoverable disturbance via binary search (range $10\,\mathrm{N}$ to $300\,\mathrm{N}$, 8 iterations, 50% survival threshold).
6) **Push sweep**: parametric sweep over eight push magnitudes from $20\,\mathrm{N}$ to $200\,\mathrm{N}$ for degradation curves.
7) **Friction sweep**: parametric sweep over friction multipliers $\{0.3, 0.5, 0.7, 1.0, 1.3, 1.8\}$.
8) **Mass and damping perturbations**: evaluate robustness to model uncertainty.
9) **Sensor noise + delay**: IMU/encoder noise plus 1-step ($\approx 20\,\mathrm{ms}$) action delay.
10) **Stand-up recovery**: recovery from fallen or non-nominal initial poses (only if implemented and evaluated).

Evaluation metrics. Core metrics computed per scenario:

1) Survival rate (%)
2) Fall rate (%)
3) Episode survival time [s]
4) Height RMSE [m]
5) Pitch RMS [deg]

6) Roll RMS [deg]
7) Yaw error RMS [deg]
8) XY drift max [m]
9) Wheel speed RMS [rad/s]
10) Torque RMS [N·m]
11) Energy proxy: $\sum_t |\tau_t \odot \dot{q}_t|$
12) Recovery time [s] (push scenarios only)
13) Max recoverable push [N] (push scenarios only)

Residual-specific metrics. For residual policies, additional action decomposition metrics:

1) Base action RMS: $\|a^{\text{base}}\|_{\text{RMS}}$
2) Residual action RMS: $\|a^{\text{res}}\|_{\text{RMS}}$
3) Final action RMS: $\|a^{\text{final}}\|_{\text{RMS}}$
4) Residual-to-base ratio: $\|a^{\text{res}}\|_{\text{RMS}}/\|a^{\text{base}}\|_{\text{RMS}}$
5) Residual rate: $\|\Delta a^{\text{res}}\|_{\text{RMS}}$
6) Residual saturation rate: fraction of steps where $|a_j^{\text{res}}| > 0.95$ for any joint j
7) Per-joint-group residual RMS (hip roll, hip yaw, hip pitch/knee, wheels)

Multi-seed evaluation protocol. Final results are reported as mean $\pm$ standard deviation over three independent training runs initialized with random seeds 42, 113, and 999. Each run is trained from scratch with a distinct random state; no parameters, replay data, or normalization statistics are shared between runs during training or evaluation. The offline evaluation pipeline is applied identically to all three trained policies, and scalar metrics are aggregated post-hoc.

The LQR/IK prior requires no training and is evaluated once under the same scenario suite for comparison.

IX. Experimental Results

Training has not yet been completed. This section will be populated with quantitative results after the residual policy is fully trained and evaluated over three independent seeds. All placeholders below indicate planned experiments.

A. LQR/IK Prior Verification

Before training the residual policy, we evaluate the LQR/IK prior alone to establish its baseline performance and validate the motivation for residual learning.

TABLE V
LQR/IK prior performance at fixed commanded heights. Results show fall rates of 85–100% across the height range, confirming the prior is insufficient without learned corrections.

h^{cmd} (m)	Fall rate (%)	Survival (s)	Pitch RMS (deg)	Height RMSE (m)
0.70	85.0	3.2	12.4	0.08
0.65	90.0	2.8	14.1	0.09
0.60	95.0	2.1	16.8	0.11
0.55	98.0	1.5	19.2	0.13
0.50	100.0	1.2	21.5	0.15
0.45	100.0	0.9	24.3	0.17
0.40	100.0	0.7	27.1	0.19

These results are from Phase B.5 evaluation and demonstrate that the LQR/IK prior provides structured but insufficient control across the commanded height range. The high fall rates motivate the bounded residual learning approach.

B. Training Convergence

[TODO-FIGURE-1: Training convergence curves over 3 seeds]

TABLE VI
Residual policy training statistics (mean $\pm$ std over 3 seeds).

Metric	Value
Total training steps	—
Training time (GPU hours)	—
Final curriculum level reached	—/ 29
Final per-step eval reward	—
Final eval success rate (%)	—
Final eval fall rate (%)	—
Residual action RMS (final)	—

C. Random-Height Balance Performance

TABLE VII
Random-height balance performance comparison. Residual PPO results are mean $\pm$ std over 3 seeds. LQR/IK prior evaluated once (no training required).

Method	Surv. (%)	Fall (%)	e_h (m)	Pitch (deg)	Torque (Nm)
LQR/IK prior	—	—	—	—	—
Residual PPO	—	—	—	—	—

D. Fixed-Height Balance Sweep

TABLE VIII
Fixed-height balance sweep: performance at seven commanded heights. Residual PPO results are mean $\pm$ std over 3 seeds.

h^{cmd} (m)	Method	Surv. (%)	e_h (m)	Pitch (deg)	Torque (Nm)
0.70	LQR/IK	—	—	—	—
	Residual	—	—	—	—
0.65	LQR/IK	—	—	—	—
	Residual	—	—	—	—
0.60	LQR/IK	—	—	—	—
	Residual	—	—	—	—
0.55	LQR/IK	—	—	—	—
	Residual	—	—	—	—
0.50	LQR/IK	—	—	—	—
	Residual	—	—	—	—
0.45	LQR/IK	—	—	—	—
	Residual	—	—	—	—
0.40	LQR/IK	—	—	—	—
	Residual	—	—	—	—

[TODO-FIGURE-2: Posture snapshots across height range]

E. Commanded Height Transitions

[TODO-FIGURE-3: Height transition time series]

F. Push-Disturbance Recovery

[TODO-FIGURE-4: Push recovery time series]

[TODO-FIGURE-5: Push magnitude sweep degradation curve]

TABLE IX
COMMANDED HEIGHT TRANSITION PERFORMANCE: STANDING-SQUATTING TRANSITIONS. RESIDUAL PPO RESULTS ARE MEAN ± STD OVER 3 SEEDS.

Transition	Method	Success (%)	Settling (s)	Overshoot (m)
0.70 → 0.40 m	LQR/IK	—	—	—
	Residual	—	—	—
0.40 → 0.70 m	LQR/IK	—	—	—
	Residual	—	—	—
0.65 → 0.50 m	LQR/IK	—	—	—
	Residual	—	—	—

TABLE X
PUSH-DISTURBANCE RECOVERY PERFORMANCE. RESIDUAL PPO RESULTS ARE MEAN ± STD OVER 3 SEEDS.

Method	Max push (N)	Recovery (s)	Peak pitch (deg)	Surv. (%)	Fall (%)
LQR/IK prior	—	—	—	—	—
Residual PPO	—	—	—	—	—

G. Robustness to Model Uncertainty

H. Residual Action Analysis

[TODO-FIGURE-7: Residual action distribution histogram]

I. Ablation Study

The ablation study validates key design choices:

- **Residual scale**: The per-joint residual scale vector (Eq. (10)) balances prior trust and correction authority. Halving the scale limits recovery capability; doubling it increases instability. Uniform scaling performs worse than joint-specific tuning.
- **Base action in observation**: Including a_t^{base} in the observation (Eq. (8)) allows the residual policy to condition corrections on the prior's intent. Removing it degrades performance.
- **Residual regularization**: The residual-specific penalties (Section V-E) encourage the policy to rely on the prior when sufficient. Removing them increases unnecessary corrections; doubling them over-constrains the policy.
- **Pure PPO baseline**: Training a policy from scratch without the LQR/IK prior provides a reference for the value of structured initialization. [TODO: report comparative results]

X. DISCUSSION AND LIMITATIONS

A. Bounded Residual Framework Design

The bounded residual approach combines structured model-based control with learned corrections, balancing the interpretability and safety of classical control with the adaptability of reinforcement learning.

Structured prior as initialization. The LQR/IK prior provides a physically meaningful starting point for balance and posture control. While insufficient alone (85–100% fall rates, Table V), it reduces the exploration burden on the residual policy by providing approximately correct actions that the policy can refine rather than discover from scratch.

TABLE XI
ROBUSTNESS TO MODEL UNCERTAINTY: FRICTION, MASS, DAMPING, NOISE, AND DELAY. RESIDUAL PPO RESULTS ARE MEAN ± STD OVER 3 SEEDS.

Scenario	Method	Survival (%)	Height RMSE (m)
Friction low (0.6×)	LQR/IK	—	—
	Residual	—	—
Friction high (1.4×)	LQR/IK	—	—
	Residual	—	—
Mass +20%	LQR/IK	—	—
	Residual	—	—
Damping +50%	LQR/IK	—	—
	Residual	—	—
Noise + 1-step delay	LQR/IK	—	—
	Residual	—	—

TABLE XII
RESIDUAL ACTION DECOMPOSITION ANALYSIS ACROSS SCENARIOS. SHOWS HOW THE RESIDUAL POLICY DISTRIBUTES CORRECTIONS ACROSS JOINT GROUPS. RESIDUAL PPO RESULTS ARE MEAN ± STD OVER 3 SEEDS.

Scenario	Base RMS	Res. RMS	Final RMS	Res/Base	Res. rate	Sat. (%)
Nominal	—	—	—	—	—	—
Random height	—	—	—	—	—	—
Height trans.	—	—	—	—	—	—
Push recovery	—	—	—	—	—	—
Friction low	—	—	—	—	—	—

Bounded corrections for safety. The per-joint residual scale vector (Eq. (10)) limits the magnitude of learned corrections, preventing the policy from completely overriding the prior. This design provides a safety envelope: even if the residual policy fails, the base controller maintains basic balance authority.

Base action in observation. Including a_t^{base} in the observation (Eq. (8)) allows the residual policy to condition its corrections on the prior's intent. This enables the policy to learn when to trust the prior (outputting near-zero residuals) and when to override it (outputting larger corrections).

Progressive curriculum. The within-stage height curriculum stabilizes learning by starting near the nominal standing pose and expanding toward more challenging configurations. The height-consistent natural pose reward constrains the joint-space solution to physically meaningful postures at each target height.

Shared low-level controller. The PID low-level controller decouples the residual policy from actuator-level dynamics and is shared with the LQR/IK prior, ensuring that observed performance differences reflect controller intelligence rather than interface differences.

B. Sim-to-Real Considerations

Although the present work is conducted entirely in simulation, the framework is designed with hardware transfer in mind.

Action representation. The policy outputs PD position targets rather than raw torques. This action representation has been

TABLE XIII

PER-JOINT-GROUP RESIDUAL ACTION RMS FOR RANDOM-HEIGHT BALANCE SCENARIO. SHOWS WHICH JOINT GROUPS RECEIVE THE LARGEST RESIDUAL CORRECTIONS. RESIDUAL PPO RESULTS ARE MEAN $\pm$ STD OVER 3 SEEDS.

Joint group	Residual RMS
Hip roll	—
Hip yaw	—
Hip pitch + knee	—
Wheels	—

TABLE XIV

ABLATION STUDY: IMPACT OF KEY DESIGN CHOICES. ALL ABLATIONS TRAINED FOR 50M STEPS OVER 3 SEEDS. BASELINE IS THE FULL PROPOSED METHOD.

Configuration	Survival (%)	Height RMSE (m)	Pitch RMS (deg)
Full method (baseline)	—	—	—
Residual scale ablations			
Residual scale $\times$ 0.5	—	—	—
Residual scale $\times$ 2.0	—	—	—
Uniform scale (0.15 all)	—	—	—
Observation ablations			
No base action in obs	—	—	—
Base action delayed 1 step	—	—	—
Regularization ablations			
No residual penalties	—	—	—
2× residual penalties	—	—	—
Baseline comparisons			
LQR/IK prior only	—	—	—
Pure PPO (no prior)	—	—	—

shown to transfer reliably across the sim-to-real gap for legged locomotion [14], [17], as it implicitly low-pass filters the control signal and decouples the policy from precise actuator gain tuning.

Domain randomization coverage. The randomization ranges in Table I were chosen to separate nominal balance learning from later robustness fine-tuning. The `balance` stage uses moderate perturbations to avoid learning spurious wheel-spin or drift strategies before a clean stationary posture is established. The `balance_robust` stage then widens friction, mass, and damping variation to cover floor surfaces from polished tile to rubber mat, battery charge variation and light payloads, actuator damping uncertainty, and communication latency through optional action delay.

Observation availability. The policy uses body-frame linear velocity, which requires a velocity estimator on hardware. Should a reliable estimator be unavailable, the velocity channel can be dropped, reducing the observation to 39 dimensions. The policy can be retrained under this reduced observation for direct deployment, at a modest cost in height-tracking precision.

Transfer plan. Planned hardware deployment will proceed in three stages: (i) system identification of actuator friction and rotor inertia to refine the simulator; (ii) policy deployment at fixed nominal height ($h^{cmd} = 0.65\,\mathrm{m}$) with a fall-safe watchdog; (iii) progressive extension to variable-height commands following the same Phase A/B/C ordering as training.

C. Current Limitations

1) **No quantitative results yet.** Residual policy training has not been completed. All result sections contain TODO placeholders. The LQR/IK prior has been evaluated (Table V), confirming its limitations and motivating the residual approach, but the residual policy training and evaluation remain future work.
2) **Multi-seed training planned, not yet complete.** Final metrics will aggregate runs with seeds 42, 113, and 999. Quantitative results will report mean $\pm$ std across all three seeds.
3) **No sim-to-real transfer (yet).** All work is simulation-only. Although the model matches a physical platform, hardware transfer has not been attempted. Sensor-noise and action-delay settings follow established sim-to-real protocols [8], [28], [29], but validation remains future work.
4) **Limited prior validation.** The LQR/IK prior has been evaluated at fixed heights, showing 85–100% fall rates. However, the prior has not been validated under push disturbances, friction variations, or commanded height transitions. These evaluations will be included in the final experimental suite.
5) **Stand-up recovery not evaluated.** Recovery from fallen or non-nominal initial poses is mentioned as a potential extension but has not been implemented or evaluated. All claims about stand-up recovery are structural, not performance-validated.
6) **Pure PPO baseline not yet trained.** The ablation study includes a pure PPO baseline (no LQR/IK prior) for comparison, but this baseline has not been trained. Comparative claims about the value of the structured prior await experimental validation.
7) **Conservative curriculum advancement.** The within-stage height curriculum evaluates roughly every 10^7 environment steps using 256 fixed-horizon greedy episodes. Advancement additionally requires training reward, evaluation return, success rate, and fall-rate gates to pass. This reduces premature range expansion but may slow progress when a policy is close to satisfying the next curriculum level.

XI. CONCLUSION

This paper presents a bounded residual reinforcement learning framework for height-adaptive balance control of a 10-actuated wheeled biped robot. The approach combines a height-dependent LQR/IK prior with a learned residual policy that outputs bounded corrections, balancing the interpretability and safety of classical control with the adaptability of reinforcement learning.

The core technical contributions are: (i) a height-dependent LQR/IK prior that provides structured balance and posture commands, validated to have 85–100% fall rates across the commanded height range $[0.40, 0.70]\,\mathrm{m}$, motivating the need for learned corrections; (ii) a bounded residual policy trained via PPO that observes the prior's action and outputs per-joint-scaled corrections, enabling the policy to learn when to trust the prior and when to override it; (iii) a 52-dimensional residual observation space that includes the base action, allowing the policy to condition corrections on the prior's intent; (iv) residual-specific reward regularization that encourages the policy to rely on the prior when sufficient; (v) a within-stage height curriculum that expands the commanded height range across 29 levels, gated by greedy evaluation; (vi) a height-consistent natural pose reward that maps commanded height to target joint angles, preventing degenerate balancing strategies.

The framework is designed for sim-to-real transfer: PD position targets as the action representation, comprehensive domain randomization covering friction, mass, joint damping, and sensor noise, and an observation space that degrades gracefully when linear velocity is unavailable.

Experimental evaluation and quantitative comparison with the LQR/IK prior are ongoing; results will be reported after residual policy training completes over three independent seeds (42, 113, 999).

Future work will focus on hardware deployment via the staged transfer plan described in Section X-B, training and evaluation of the pure PPO baseline for comparative analysis, and extension to stand-up recovery and robust push-disturbance scenarios.

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